

Can a Neuron Store Infinite Information?

Continuous Morphology, Finite Biological Resolution, and Accessible Neural Memory

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Final preprint, Version 1.0 — August 8, 2026

Abstract

A hand-drawn curve can be described at successively finer resolution. Each increase in resolution reveals distinctions that were absent from the coarser description. In 2012, Micah Blumberg used this intuition to argue that neuronal topology could support an extremely large information space and that a neuron's electrical output could provide a sparse readout of its richer structural state. The word *infinite* makes the idea memorable but also hides three different questions: how many distinctions an ideal continuous object admits, how many physical states a biological neuron can reliably occupy, and how many of those states a network can actually recover and use.

This paper separates those capacities. An ideal continuous morphology has no fixed finite description length when measurement resolution is allowed to approach zero. A biological neuron, however, has finite volume, energy, molecules, stability, observation time, and signal-to-noise ratio; its reliably distinguishable state capacity is finite under every realizable measurement protocol. That finite capacity can nevertheless be much richer than a scalar-weight description because branch topology, synapse placement, nonlinear dendritic sub-units, receptor and channel state, intracellular organization, and temporal context jointly shape the function implemented by the cell. Functional capacity is smaller still: two physical states count as different memories only when some controlled input produces a reliably distinguishable consequence. Accessible capacity is finally bounded by the output channel and downstream receiver.

We formalize resolution-dependent packing, correlated noise-limited capacity, functional equivalence classes, and the readout bottleneck. A deterministic illustration shows logarithmic growth of resolution-dependent capacity, loss of joint capacity as coordinate correlations increase, and saturation of accessible information when the readout dimension is smaller than the morphological state. The calculations are demonstrations of the bounds, not estimates of neuronal storage. The paper replaces a literal infinite-biological-storage claim with a stronger testable proposition: neuronal morphology may support a large, multiscale state space, but only the reproducibly distinguishable and causally readable fraction is memory available to a network.

Keywords: neural information capacity; dendritic morphology; structural plasticity; metric entropy; Phase Wave Differential; functional equivalence; biological memory

1. The Curve in the Neuron

Consider a curved dendritic branch. At one-millimeter resolution it has one description; at micrometer resolution it has another; at molecular resolution it contains still more distinctions. If the curve is treated as an ideal real-valued object and no minimum scale is imposed, there is no final finite list of all its possible

distinctions.

The biological question is not whether mathematics permits infinite precision. It is:

How many structurally different neuronal states can exist, remain stable, be selected by input, and produce consequences that another part of the nervous system can distinguish?

That question matters because conventional neural-network abstractions often assign one scalar weight to each connection. Biological memory can also involve where a contact is located, which contacts cooperate within a branch, local receptor and channel distributions, branch excitability, intracellular organization, and how these variables change across time. The possible state space may be enormous even though it is not literally infinite under physical conditions.

The solution begins by separating three capacities that are often collapsed into one word.

2. Three Capacities, Not One

2.1 Ideal description capacity

An idealized continuous object admits arbitrarily fine distinctions. Its required description length grows as the declared resolution becomes finer. This is a mathematical statement about a model class, not a claim about a cell.

2.2 Physical distinguishability capacity

A biological neuron is finite, noisy, metabolically constrained, and changing. A physical state is meaningful only relative to an observation duration, error tolerance, and measurement scale. At every finite resolution and tolerance, only finitely many states can be reliably distinguished.

2.3 Functional and accessible capacity

Not every physical difference changes what the neuron does. Two arbors can differ microscopically yet implement the same measurable input-output function. Even a functionally relevant difference may fail to pass through the cell's output or may be ignored by a downstream receiver. The memory available to a network is therefore bounded by the narrowest stage in the chain:

physical state

- > input-selectable function
- > reliable cellular output
- > receiver discrimination
- > task-relevant consequence

3. Historical Genealogy and the Modern Question

The September 2012 SAN source described a curve as capable of containing arbitrarily many distinctions and proposed that neuronal topology could function as a program. Companion records described an electrical packet as a sparse representation of the wider dendritic and cellular state. Those records establish the ancestry of the capacity argument. They do not contain the later PWD acronym, a channel-capacity theorem, or a claim that an actual neuron can recover infinitely many physical bits.

The public SAN Git history fixed the relevant notes in 2022 and added the mature receiver-relative Phase Wave Differential concept. This creates a disciplined update:

2012: continuous curve and morphology-as-program intuition

-> 2012: sparse electrical readout of a richer cellular state

-> 2022: typed receiver-relative output difference

-> 2026: resolution, noise, functional-equivalence, and readout bounds

The modern paper preserves the central intuition while identifying the conditions under which it becomes finite, measurable, and falsifiable.

4. Resolution-Dependent State Counting

Let the effective neuronal state be represented by d coordinates over a bounded set $\mathcal{S}$. Coordinate j spans a declared range R_j and can be distinguished only to resolution $\epsilon_j > 0$. A simple axis-aligned packing bound is

$$C_{\epsilon_1, \dots, \epsilon_d} \leq \sum_{j=1}^d \log_2 \left(1 + \frac{R_j}{\epsilon_j} \right). \quad (1)$$

For every finite d and every positive ϵ_j , the bound is finite. In the ideal limit $\epsilon_j \rightarrow 0$, it grows without a fixed upper value. This is the precise mathematical content of the curve intuition: unbounded description capacity appears only when arbitrarily fine precision is permitted.

A more general formulation uses the covering number $N(\mathcal{S}, d_{\mathcal{S}}, \epsilon)$, the minimum number of radius- ϵ neighborhoods needed to cover the state space under a declared metric $d_{\mathcal{S}}$:

$$H_{\epsilon}(\mathcal{S}) = \log_2 N(\mathcal{S}, d_{\mathcal{S}}, \epsilon). \quad (2)$$

Equation 2 makes the dependence on the metric explicit. A morphology measured only by total length has a different capacity from one measured by branch topology, synapse placement, receptor state, and time-varying excitability. Capacity is therefore never one context-free number.

5. Noise, Correlation, and Stability

If a measurement vector $\mathbf{x}$ has signal covariance $\mathbf{S}$ and noise covariance $\mathbf{N}$, a Gaussian linear-channel comparator is

$$C_G = \frac{1}{2} \log_2 \det(\mathbf{I} + \mathbf{N}^{-1/2} \mathbf{S} \mathbf{N}^{-1/2}). \quad (3)$$

This is an upper-bound comparator under Gaussian assumptions, not a declaration that morphology is an independent Gaussian communication channel. Correlation reduces the number of independently usable directions, and nonlinear sampling can either reveal or hide functional distinctions.

Memory must also persist over a declared interval τ . Let $P_e(\tau)$ be the minimum achievable classification error after state drift and turnover. An operationally stable capacity can be defined as

$$C_{\text{stable}}(\tau, \alpha) = \log_2 M^*, \quad M^* = \max \{M : P_e(\tau) \leq \alpha\}, \quad (4)$$

where α is a predeclared error tolerance. This definition prevents a transient molecular distinction from being counted as long-term memory unless the distinction can be recovered after the relevant delay.

6. Functional Equivalence Classes

Physical difference is not sufficient. Let $F(\mathbf{u}, \mathbf{s}, \mathbf{h})$ be the output produced from input $\mathbf{u}$, cellular state $\mathbf{s}$, and recent history $\mathbf{h}$. Two states are functionally equivalent under an input set $\mathcal{U}$ and tolerance δ when

$$\mathbf{s}_a \equiv_{F, \mathcal{U}, \delta} \mathbf{s}_b \iff \sup_{\mathbf{u} \in \mathcal{U}} D(F(\mathbf{u}, \mathbf{s}_a, \mathbf{h}), F(\mathbf{u}, \mathbf{s}_b, \mathbf{h})) \leq \delta. \quad (5)$$

The number of functional memory states is the number of equivalence classes,

$$C_{\text{functional}} = \log_2 |\mathcal{S} / \equiv_{F, \mathcal{U}, \delta}|. \quad (6)$$

This formulation captures the useful part of the 2012 statement that topology is a program. Morphology matters when it changes the transform performed on a relevant set of inputs. It does not gain functional bits merely by containing unmeasured detail.

7. From Structural Capacity to an Accessible Readout

Suppose an input selects a morphology-dependent function and evokes a typed output $\mathbf{Y}$. A downstream receiver observes $\mathbf{Z}$. By the data-processing inequality for the declared chain $\mathbf{S} \rightarrow \mathbf{Y} \rightarrow \mathbf{Z}$,

$$I(\mathbf{S}; \mathbf{Z} \mid \mathbf{U}, \mathbf{H}) \leq I(\mathbf{S}; \mathbf{Y} \mid \mathbf{U}, \mathbf{H}) \leq H(\mathbf{S}). \quad (7)$$

The accessible capacity is therefore bounded by

$$C_{\text{accessible}} \leq \min(C_{\text{stable}}, C_{\text{functional}}, C_{\text{output}}, C_{\text{receiver}}). \quad (8)$$

The output may include a receiver-relative difference in phase, latency, duration, waveform, burst state, or release consequences. Once that operation has been defined, SAN calls the typed difference a **Phase Wave Differential**. PWD is relevant here because it supplies a candidate readout bottleneck. A morphology may admit many stable distinctions while one PWD exposes only the task-relevant projection that a receiver can use.

7.1 Decoding gives a lower bound, not the whole capacity

Suppose an experiment defines M equally likely state classes and a held-out decoder has error probability P_e . Fano's inequality gives

$$I(\mathbf{S}; \mathbf{Z}) \geq \log_2 M - h_2(P_e) - P_e \log_2(M - 1), \quad (9)$$

where h_2 is binary entropy. This converts classification into a conservative accessible-information estimate under the declared class design. A poor decoder cannot prove that information is absent, because the decoder

family or measurement may be inadequate; a successful decoder establishes only that at least that much class information is recoverable.

The final task-relevant quantity should include intervention. Let Z be the receiver response and B the behavioral or network consequence. Define

$$C_{\text{usable}} \leq \min(C_{\text{accessible}}, I(\mathbf{S}; B | \mathbf{U})), \quad (10)$$

and require a preregistered perturbation to show that changing the decoded channel changes B . This separates readable correlation from usable memory.

8. Why Morphology Could Still Matter Greatly

Finite does not mean small. A scalar weight model assigns a value to a connection. A biological model can additionally depend on branch identity, contact location, local nonlinear integration, receptor and conductance state, spine clustering, and the temporal ordering of input. Poirazi and Mel’s analysis showed that nonlinear dendritic subunits and structural plasticity can yield substantially greater capacity than linear integration. Losonczy and colleagues demonstrated branch-specific storage of correlated input features. Learning-related spine clustering and branch-specific spine formation provide in vivo evidence that structural change can be organized at a subcellular scale. Aizenbud and colleagues’ Functional Complexity Index provides a recent way to quantify how morphology and NMDA nonlinearity contribute to the diversity of input-output functions in detailed neuronal models.

These studies establish neither literal infinity nor the PWD readout. They make the central empirical question serious: how much additional held-out functional discrimination is gained when measured structure and biophysics are added to a scalar synapse-and-rate model?

9. Deterministic Capacity Demonstration

9.1 Design

The accompanying script illustrates three mathematical effects with fixed parameters and a fixed seed. First, a 12-coordinate bounded state is measured at progressively finer resolutions. Second, a correlated Gaussian comparator shows how redundancy reduces joint capacity. Third, a lower-dimensional noisy readout demonstrates that accessible capacity saturates below the full structural state.

The model is not fitted to neuronal data. Its numbers are dimensionless and illustrative. Its purpose is to make the logic executable and expose any contradiction among the proposed bounds.

9.2 Results

For five resolutions—0.2, 0.1, 0.05, 0.025, and 0.0125—the packing illustration yielded 31.51, 42.87, 54.45, 66.26, and 78.17 bits, respectively. The increase is monotonic but finite at every nonzero resolution.

For a fixed signal-to-noise setting and correlations $\rho = 0, 0.25, 0.50, 0.75$, the Gaussian comparator yielded 15.04, 14.08, 12.12, 8.99 bits. Increasing correlation reduced independent capacity.

When the readout exposed 1, 2, 4, 8, 12 effective dimensions, the intermediate PWD-channel comparator yielded 1.44, 2.52, 4.61, 8.98, 13.40 bits, while the more restrictive receiver channel yielded accessible val-

ues of 1.11, 1.89, 3.42, 6.66, 9.98 bits. The result shows the bottleneck directly: a rich state does not become available merely because it exists.

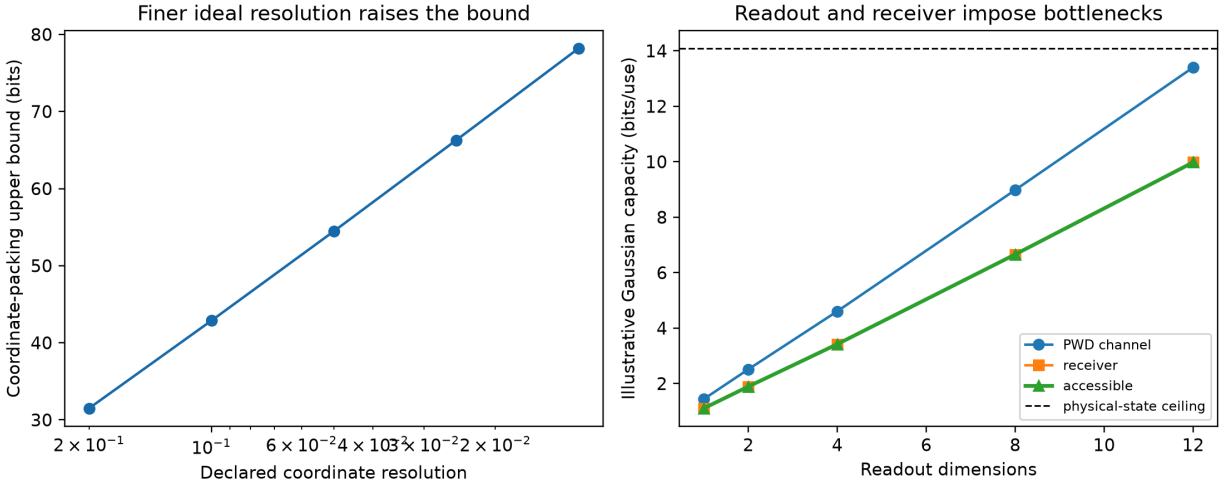


Figure 1: Resolution, correlation, and readout capacity demonstration

10. An Empirical Measurement Program

10.1 Declare the state and scale

Choose a cell type, learning task, retention interval, and state variables before measuring capacity. Reconstruct morphology and synapse placement at a declared resolution; measure selected receptor, conductance, and excitability variables; and report missing dimensions rather than treating the reconstruction as complete.

10.2 Estimate physical distinguishability

Repeat measurements to estimate noise, drift, and reconstruction error. Use held-out sessions to determine which state classes remain distinguishable after the retention interval. Report capacity as a curve over spatial resolution, temporal resolution, and error tolerance.

10.3 Estimate functional equivalence

Deliver a prespecified family of spatial-temporal inputs. Fit scalar-weight, nonlinear-branch, and richer biophysical models on training trials and compare their held-out predictions. Cluster physical states by functional equivalence rather than counting every anatomical difference.

10.4 Estimate output and receiver information

Measure spike rate, event timing, phase, waveform, duration, burst structure, and release-related observables where technically possible. Quantify conditional information about structural state after input and recent history are controlled. Then measure how much survives at identified receivers.

10.5 Establish causality

Perturb a predicted structural feature, measure the loss in function and readout, and perform matched rescue. Correlation between a large arbor and complex output is insufficient if a different hidden variable explains both.

10.6 Correct finite-sample bias

Capacity estimates rise spuriously when model flexibility is large relative to sample size. Use nested cross-validation, held-out cells and animals, permutation-derived nulls, estimator-bias correction, and confidence intervals over biological units. Report the sensitivity of every estimate to state-space discretization, decoder family, and the chosen metric. A single in-sample bit count is not a capacity measurement.

11. Comparator Hierarchy

The empirical result should be reported against increasingly strong alternatives:

1. total synapse count;
2. scalar synaptic weights;
3. firing rate and spike count;
4. nonlinear dendritic subunits without detailed morphology;
5. morphology plus conductance and receptor state;
6. morphology plus recent temporal state; and
7. morphology plus typed output and receiver response.

The proposed SAN increment is supported only when the later stages add reproducible held-out information and causal control. A large parameter count alone is not evidence of a large memory capacity because correlated, unstable, or unreadable parameters may contribute little.

12. Failure Conditions

The strong morphology-capacity proposition is weakened if:

- added structural variables do not improve held-out functional prediction;
- apparent capacity disappears after reconstruction error and state drift are included;
- structural distinctions cannot be selected by controlled inputs;
- output features contain no structural-state information beyond mean rate and recent history;
- receiver responses do not preserve the structural distinction; or
- causal perturbation and rescue do not behave as predicted.

The ideal mathematical result remains true even if the biological hypothesis fails: continuous models admit finer descriptions as resolution improves. The scientific value lies in measuring where biology, noise, stability, function, and readout impose the actual bounds.

13. Discussion

The question in the title has a conditional answer. An ideal continuous neuron model can require unbounded description as precision approaches infinity. A physical neuron cannot provide infinite recoverable information under finite resources and finite observation. Yet its functional state space can still be far larger and more structured than a scalar abstraction suggests.

The Bekenstein bound supplies a distant ceiling for entropy in a bounded physical system, but it is not a useful estimate of neuronal memory. It ignores the much tighter biological limits imposed by temperature, noise, molecular turnover, metabolic cost, state accessibility, measurement, and readout. The operative bounds in this paper are therefore resolution-, stability-, function-, and receiver-dependent.

This distinction strengthens the original 2012 argument. It turns *infinite* from a literal biological quantity into a limit statement and a research program. The important questions become: which dimensions matter, how stable are they, what inputs select them, what output exposes them, and what receiver can use them? Those questions can be answered experimentally.

The result also clarifies the relationship among the three papers in this sequence. The present paper defines the capacity landscape. *From Morphology to Phase* specifies how a learned state can be evoked as a typed local readout. *From Cellular Difference to Population Reinstatement* asks how that local readout can survive transformation across larger populations.

14. Conclusion

A neuron need not store infinite physical information for morphology to be a major memory substrate. The defensible claim is both more modest and more consequential: neuronal structure and biophysics may provide a high-dimensional, multiscale set of stable functions, while output and receiver dynamics expose only a sparse accessible projection. Measuring the gap among ideal, physical, functional, and accessible capacity is the path from a provocative curve analogy to a testable theory of neural memory.

Research Ethics and Data Scope

This theoretical paper reports no new human or animal experiment. Future capacity measurements must use appropriate institutional review and welfare protections, disclose reconstruction uncertainty and exclusions, and distinguish repeated technical samples from independent biological units. The numerical values reported here are generated by the included deterministic illustrative model and are not measurements of neuronal storage.

Code, Data, and Reproducibility

The deterministic illustration, machine-readable results, and figure are included with this manuscript:

- `model/run_capacity_bounds_demo.py`
- `model/results/capacity_bounds_demo_results.json`
- `model/results/capacity_bounds_demo.csv`
- `output/figures/capacity-resolution-and-readout-bounds.png`

All numerical values are illustrative and should not be interpreted as measured neuronal bits.

Authorial Provenance and AI Assistance

The curve, morphology-as-program, and sparse-readout ancestry is preserved in Micah Blumberg’s 2012 source record. The mature PWD readout concept appears in the 2022 SAN corpus. AI tools assisted source recovery, formalization, model implementation, and editorial review. The historical theory and authorship belong to Micah Blumberg.

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